

Task 3: Installation of Librem5 and Pure OS

1. Introduction

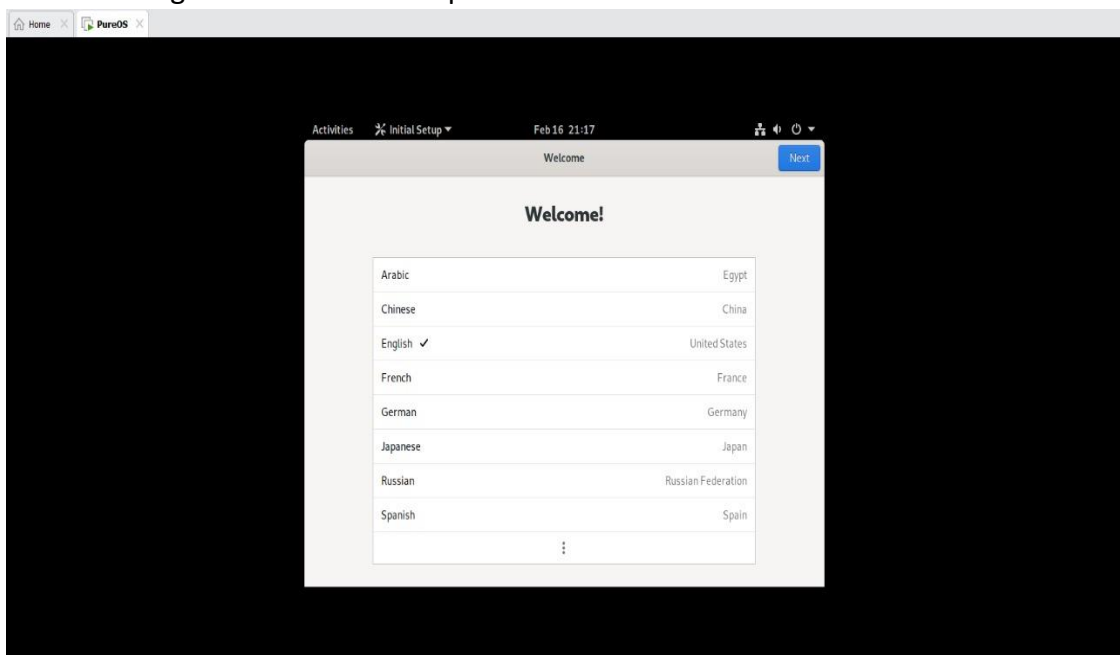
- Librem 5 is a privacy-focused smartphone developed by Purism, designed to prioritize security, user control, and open-source software.
- It runs on PureOS, a Debian-based operating system built for privacy and freedom.

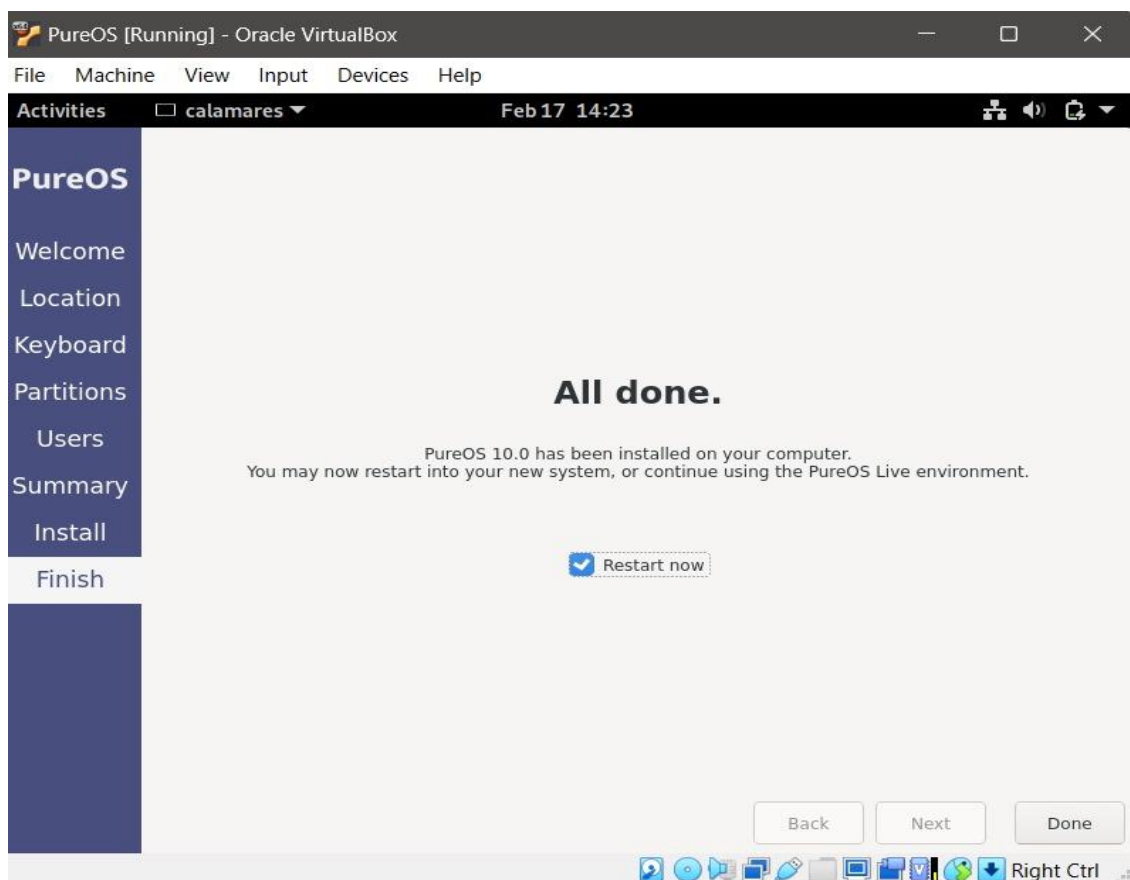
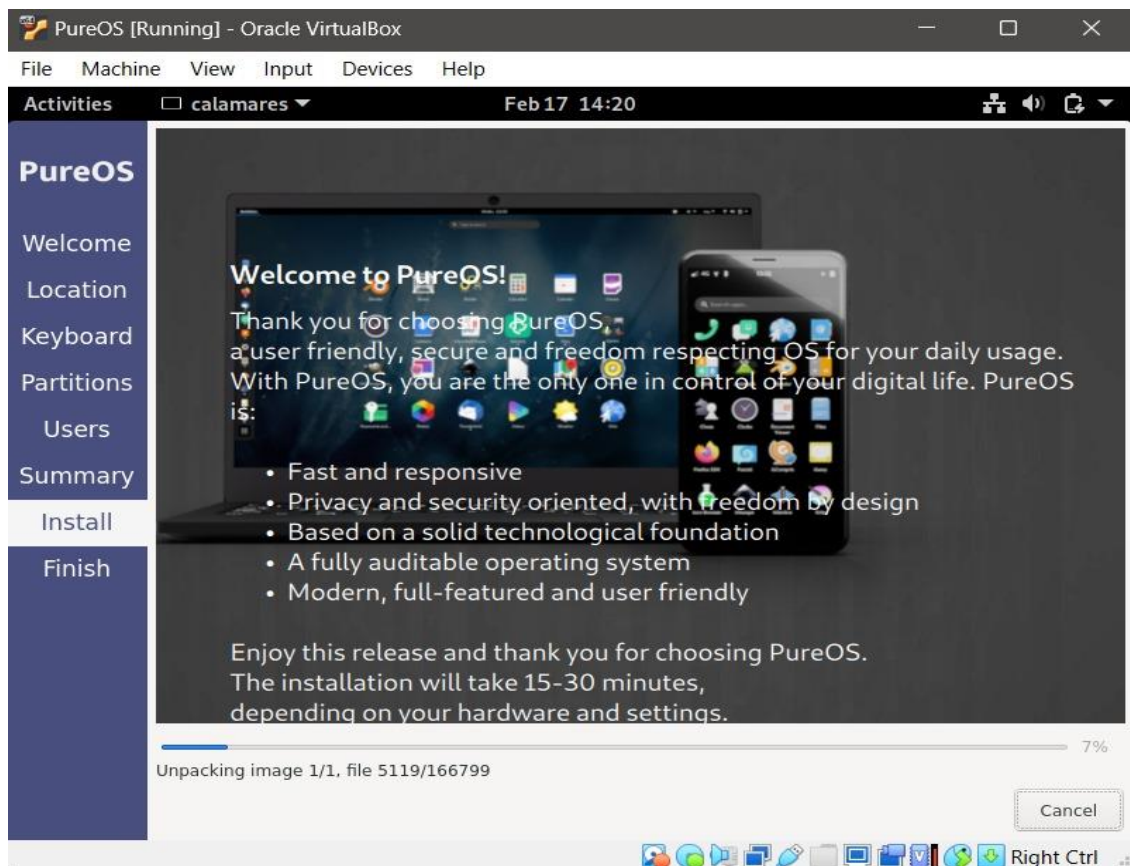
2. Steps for Installation:

2.1 Download the ISO file and launch PureOS



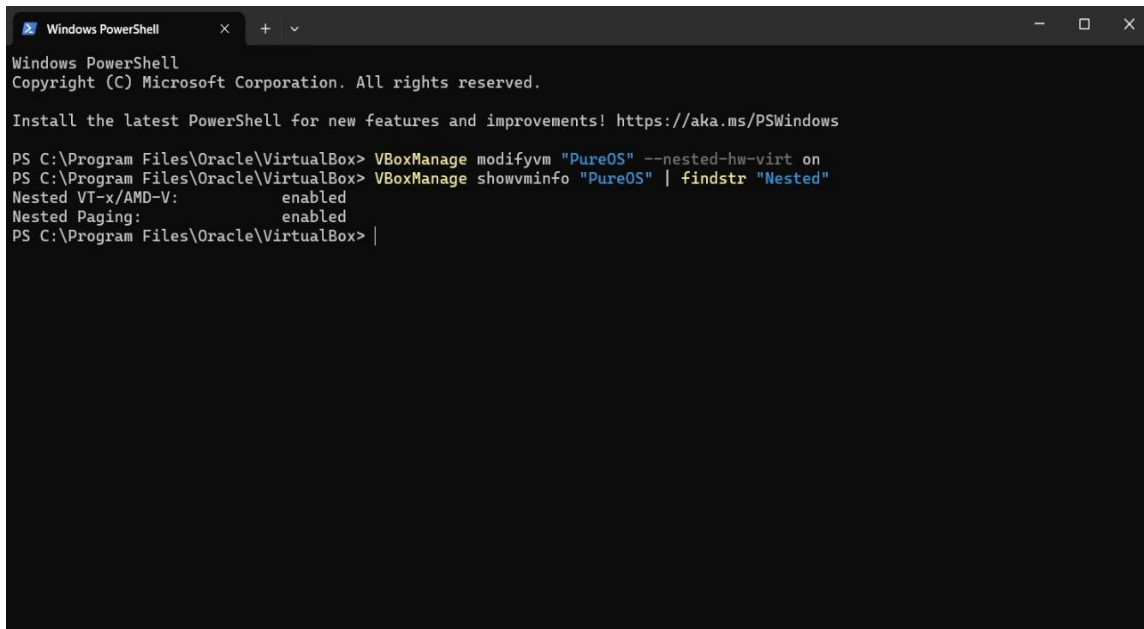
3.2 Go through all the installation process





3.3 Open Command Prompt in your VirtualBox Installation Folder and run the following commands:

- "VBoxManage modifyvm "PureOS_VM_Name_In_VirtualBox" --nested-hw-virt on"
- "VBoxManage showvminfo "PureOS_VM_Name_In_VirtualBox" | findstr "Nested""



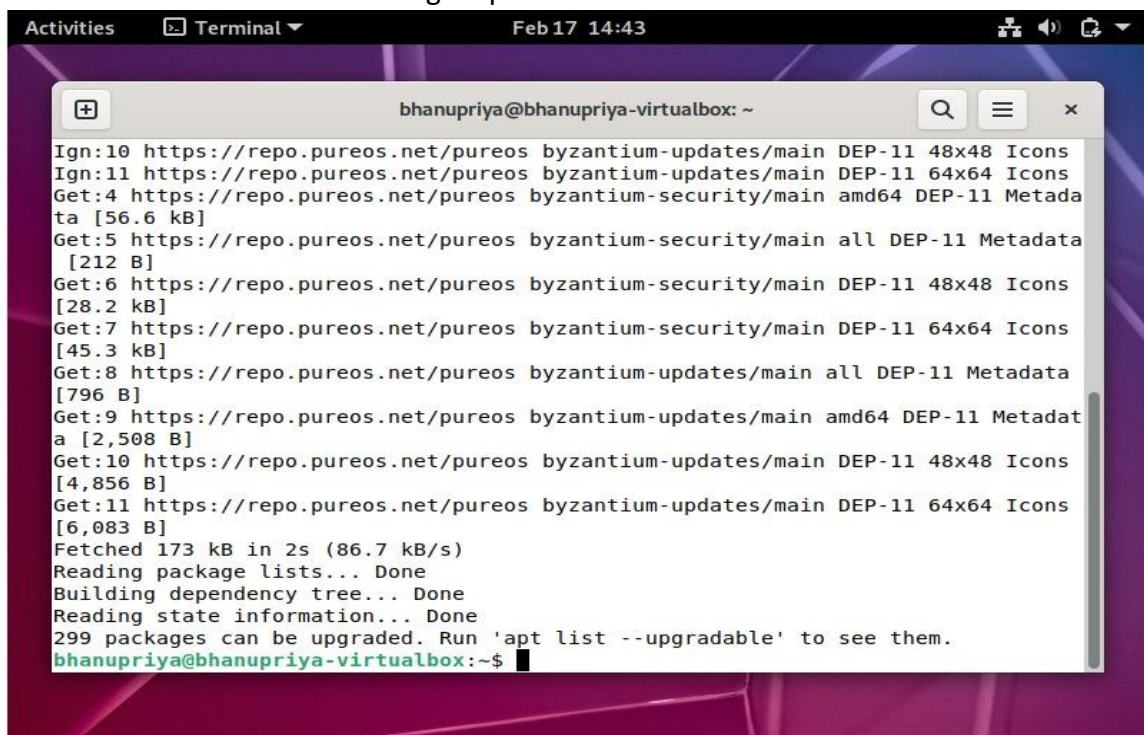
```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Program Files\Oracle\VirtualBox> VBoxManage modifyvm "PureOS" --nested-hw-virt on
PS C:\Program Files\Oracle\VirtualBox> VBoxManage showvminfo "PureOS" | findstr "Nested"
Nested VT-x/AMD-V:          enabled
Nested Paging:              enabled
PS C:\Program Files\Oracle\VirtualBox> |
```

3.4 Open the Terminal and run the following command

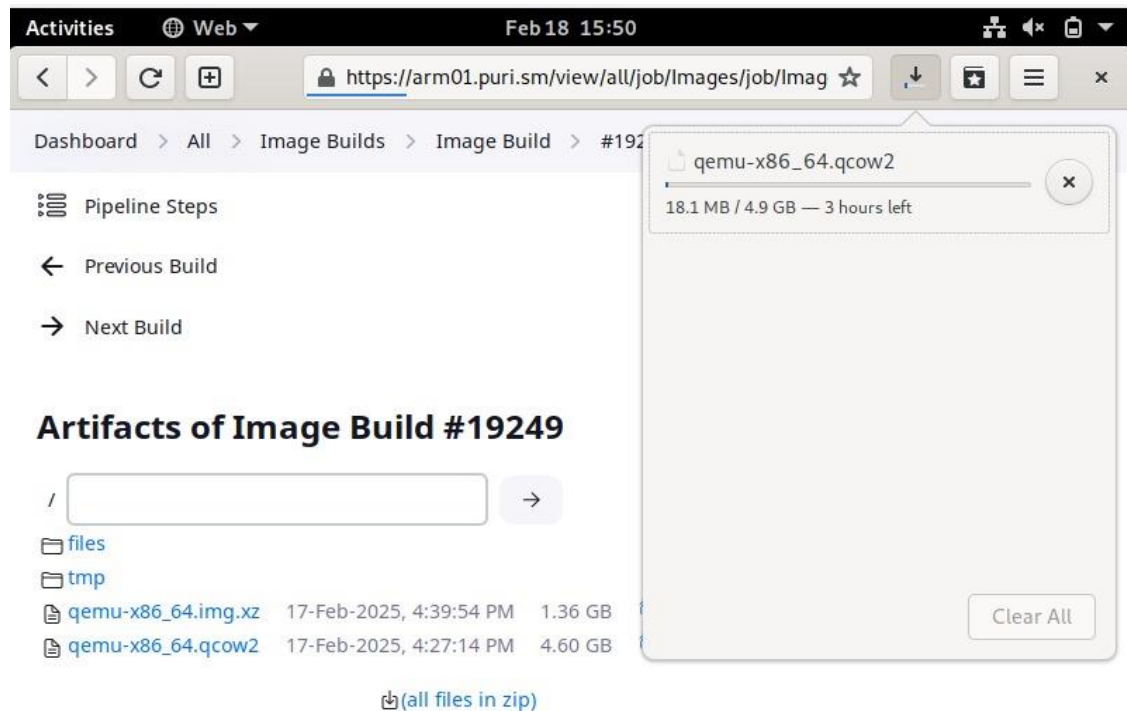
- "sudo apt update"
- "sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils libvirt-daemon virt-manager cpu-checker"



```
Activities  Terminal  Feb 17 14:43

bhanupriya@bhanupriya-virtualbox: ~
Ign:10 https://repo.pureos.net/pureos byzantium-updates/main DEP-11 48x48 Icons
Ign:11 https://repo.pureos.net/pureos byzantium-updates/main DEP-11 64x64 Icons
Get:4 https://repo.pureos.net/pureos byzantium-security/main amd64 DEP-11 Metadata [56.6 kB]
Get:5 https://repo.pureos.net/pureos byzantium-security/main all DEP-11 Metadata [212 B]
Get:6 https://repo.pureos.net/pureos byzantium-security/main DEP-11 48x48 Icons [28.2 kB]
Get:7 https://repo.pureos.net/pureos byzantium-security/main DEP-11 64x64 Icons [45.3 kB]
Get:8 https://repo.pureos.net/pureos byzantium-updates/main all DEP-11 Metadata [796 B]
Get:9 https://repo.pureos.net/pureos byzantium-updates/main amd64 DEP-11 Metadata [2,508 B]
Get:10 https://repo.pureos.net/pureos byzantium-updates/main DEP-11 48x48 Icons [4,856 B]
Get:11 https://repo.pureos.net/pureos byzantium-updates/main DEP-11 64x64 Icons [6,083 B]
Fetched 173 kB in 2s (86.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
299 packages can be upgraded. Run 'apt list --upgradable' to see them.
bhanupriya@bhanupriya-virtualbox:~$
```

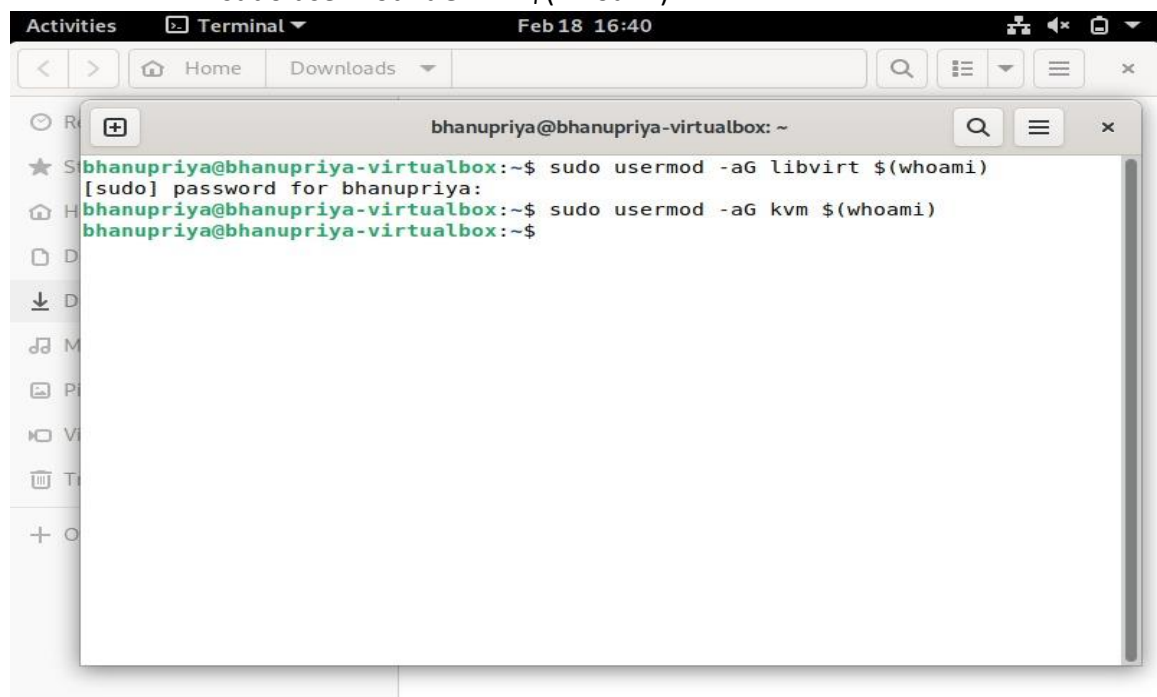
3.4 Open the Web Application and go the website (<https://arm01.puri.sm/view/all/job/Images/job/Image%20Build/>) and download the .qcow2 file from the image build number 19249.



Jenkins 2.426.3

3.5 In the terminal, run the following commands:

- "sudo usermod -aG libvirt \$(whoami)"
- "sudo usermod -aG kvm \$(whoami)"



3.6 Then, log out and log back in to the Pure OS Account to make the changes then run the following commands:

- "sudo systemctl enable libvirtd"
- "sudo systemctl start libvirtd"
- Check whether libvirtd is running and is active with the command:
"sudo systemctl status libvirtd"

```
bhanupriya@bhanupriya-virtualbox: ~  
bhanupriya@bhanupriya-virtualbox:~$ sudo systemctl enable libvirtd  
[sudo] password for bhanupriya:  
bhanupriya@bhanupriya-virtualbox:~$ sudo systemctl start libvirtd  
bhanupriya@bhanupriya-virtualbox:~$ sudo systemctl status libvirtd  
● libvirtd.service - Virtualization daemon  
   Loaded: loaded (/lib/systemd/system/libvirtd.service; enabled; vendor preset: enabled)  
   Active: active (running) since Tue 2025-02-18 16:08:51 IST; 35min ago  
     TriggeredBy: ● libvirtd-admin.socket  
                  ● libvirtd.socket  
                  ● libvirtd-ro.socket  
        Docs: man:libvirtd(8)  
              https://libvirt.org  
    Main PID: 649 (libvirtd)  
      Tasks: 19 (limit: 32768)  
    Memory: 23.8M  
       CPU: 155ms  
    CGroup: /system.slice/libvirtd.service  
            └─649 /usr/sbin/libvirtd  
  
Feb 18 16:08:51 bhanupriya-virtualbox systemd[1]: Starting Virtualization daemon:  
Feb 18 16:08:51 bhanupriya-virtualbox systemd[1]: Started Virtualization daemon.  
lines 1-17/17 (END)
```

3.8 Then click on the Applications Tab on the top left and then search for "Virtual Machine Manager" and launch it. Once you reach the Lock Screen, you need to use your mouse to pull the screen up to get to the keypad and once at the keypad, the password is 123456 and you can use your keyboard for entering password and press enter on your keyboard to unlock the phone.

